



Homework Assignment

Week 7: Introduction to Lists

Total Points: 100 — Due: Next Class



Name: _____

Date: _____

Important Instructions

- Work on your own, but you may ask family members for hints only if stuck
- Write your code in Python (*using Thonny or any online IDE*)
- For setting up your code files, kindly see `instructions.pdf`
- For written answers, use the spaces provided on this sheet
- Show your work for full credit!

1 Part 1: Understanding Lists (15 points)

Time estimate: 15 minutes

1.1 Exercise 1.1: List Basics Table (8 points)

Fill in the table to show your understanding of list indexing and operations:

Code	Output	Valid?	Explanation
<code>fruits = ["apple", "banana", "mango"]</code>		Yes	
<code>print(fruits[0])</code>			
<code>print(fruits[3])</code>			
<code>print(fruits[-1])</code>			
<code>print(len(fruits))</code>			

1.2 Exercise 1.2: List vs Individual Variables (7 points)

Circle the better approach for each scenario and explain why:

Scenario 1: Storing test scores for 20 students

- Option A: `score1 = 85, score2 = 92, score3 = 78...` (20 variables)
- Option B: `scores = [85, 92, 78, ...]` (one list)

Better choice: _____

Why? _____

Scenario 2: Storing a student's name, age, and grade

- Option A: `name = "Ali", age = 13, grade = 8` (three variables)
- Option B: `student = ["Ali", 13, 8]` (one list)

Better choice: _____

Why? _____

2 Part 2: Code Completion - List Methods (20 points)

Time estimate: 25 minutes

2.1 Exercise 2.1: Shopping List Manager (10 points)

Open [week07_ex2.1.py](#) and complete the following code that manages a shopping list:

```

1  # Shopping List Manager
2  print("=== My Shopping List ===")
3
4  # Start with an empty shopping list
5  shopping_list = _____
6
7  # Add items to the list
8  shopping_list._____ ("milk")
9  shopping_list._____ ("bread")
10 shopping_list._____ ("eggs")
11
12 print(f"Current list: {shopping_list}")
13 print(f"Number of items: {_____}")
14
15 # Insert butter at position 1 (second item)
16 shopping_list._____ (1, "butter")
17 print(f"After adding butter: {shopping_list}")
18
19 # Remove eggs from the list
20 shopping_list._____ ("eggs")
21 print(f"After removing eggs: {shopping_list}")
22
23 # Remove the last item and save it
24 last_item = shopping_list._____ ()
25 print(f"Removed last item: {last_item}")
26 print(f"Final list: {shopping_list}")

```

2.2 Exercise 2.2: Grade Tracker (10 points)

Open [week07_ex2.2.py](#) and complete this grade analysis program:

```

1  # Grade Analysis System
2  grades = [85, 92, 78, 92, 88, 95, 85]

```

```

3
4 print("== Grade Analysis ==")
5 print(f"All grades: {grades}")
6
7 # Find total number of grades
8 total_grades = _____
9 print(f"Total grades: {total_grades}")
10
11 # Calculate sum and average
12 total_score = _____(grades)
13 average = total_score / _____
14 print(f"Average score: {average:.1f}")
15
16 # Find highest and lowest scores
17 highest = _____(grades)
18 lowest = _____(grades)
19 print(f"Highest: {highest}, Lowest: {lowest}")
20
21 # Count how many times 92 appears
22 count_92 = grades._____(92)
23 print(f"Number of students with 92: {count_92}")
24
25 # Check if 100 is in the grades
26 if _____ in grades:
27     print("Someone scored 100!")
28 else:
29     print("No perfect scores yet.")

```

3 Part 3: Code Tracing (15 points)

Time estimate: 20 minutes

3.1 Exercise 3.1: Trace List Operations (8 points)

Trace through this code and write what the list looks like after each operation:

```

1 numbers = [10, 20, 30]
2 print("Step 1:", numbers)
3
4 numbers.append(40)
5 print("Step 2:", numbers)
6
7 numbers.insert(1, 15)
8 print("Step 3:", numbers)
9
10 numbers.remove(30)
11 print("Step 4:", numbers)
12
13 numbers.pop(0)
14 print("Step 5:", numbers)

```

Your answers:

After Step 1: _____

After Step 2: _____

After Step 3: _____

After Step 4: _____

After Step 5: _____

3.2 Exercise 3.2: Output Prediction (7 points)

What will this program print? Trace through it mentally:

```
1 colors = ["red", "blue", "green"]
2 colors.append("yellow")
3 print(colors[1])
4 print(colors[-1])
5 print(len(colors))
```

Line 1 output: _____

Line 2 output: _____

Line 3 output: _____

4 Part 4: Building Lists with Loops (18 points)

Time estimate: 25 minutes

4.1 Exercise 4.1: Collecting User Input (10 points)

Open [week07_ex4.1.py](#) and complete this program that collects names from the user:

```
1 # Name Collector
2 print("=== Class Name Collector ===")
3
4 # Start with empty list
5 names = _____
6
7 # Ask how many names to collect
8 num_names = int(input("How many names? "))
9
10 # Loop to collect names
11 for i in range(_____):
12     name = input(f"Enter name {i+1}: ")
13     names._____ (name)
14
15 # Display all names
16 print("\n=== Class List ===")
17 for _____ in _____:
18     print(f"- {name}")
19
20 print(f"\nTotal students: {_____}")
```

4.2 Exercise 4.2: Filtering Numbers (8 points)

Open [week07_ex4.2.py](#) and complete this program that filters numbers:

```
1 # Number Filter
2 numbers = [5, 12, 8, 20, 3, 15, 7, 18]
```

```

3
4 print(f"Original numbers: {numbers}")
5
6 # Create new list with only numbers > 10
7 big_numbers = []
8
9 for num in numbers:
10     if num > 10:
11         big_numbers.append(num)
12
13 print(f"Numbers greater than 10: {big_numbers}")
14
15 # Count how many big numbers we found
16 count = len(big_numbers)
17 print(f"Found {count} numbers greater than 10")

```

5 Part 5: Debugging Detective (12 points)

Time estimate: 20 minutes

Each code snippet has errors. Find and fix ALL errors:

5.1 Snippet 1: List Creation Error (6 points)

```

1 # This code has 3 errors - fix them all!
2 my_list = (1, 2, 3, 4, 5)      # Wrong brackets!
3 print(my_list[5])              # Index problem!
4 my_list.add(6)                 # Wrong method name!

```

Fixed version:

```

1 # Write the corrected code here:

```

Save your corrected code as week07_ex5_1_fixed.py

5.2 Snippet 2: Method Confusion (6 points)

```

1 # This code has 3 errors - fix them all!
2 fruits = ["apple", "banana", "mango"]
3 fruits.append("orange", "grape") # append takes only 1 argument!
4 fruits.delete("banana")          # Wrong method!
5 position = fruits.find("mango")  # Wrong method for lists!

```

Fixed version:

```

1 # Write the corrected code here:

```

Save your corrected code as week07_ex5_2_fixed.py

6 Part 6: Full Program Development (15 points)

Time estimate: 30 minutes

6.1 Exercise 6.1: High Score Tracker (15 points)

Create a program called `high_score_tracker.py` that manages game scores:

Program Requirements

Your program should:

- Start with an empty list for scores
- Ask the user how many scores to enter
- Collect scores from the user using a loop
- Calculate and display:
 - All scores
 - Average score
 - Highest score
 - Lowest score
 - Number of scores above 80
- Handle invalid input gracefully (no crashes!)

Example Run:

```

=== High Score Tracker ===
How many scores to enter? 4
Enter score 1: 85
Enter score 2: 92
Enter score 3: 78
Enter score 4: 95

=== Score Analysis ===
All scores: [85, 92, 78, 95]
Average: 87.5
Highest: 95
Lowest: 78
Scores above 80: 3
    
```

Starter Structure:

```

1  # high_score_tracker.py
2  # Name: -----
3  # Date: -----
4
5  print("=== High Score Tracker ===")
6
7  # Initialize empty scores list
8  scores = []
9
10 # Get number of scores
11 num_scores = int(input("How many scores to enter? "))
12
13 # Collect scores with loop
14 for i in range(num_scores):
15     # Your code here
16     pass
    
```

```

17
18 # Calculate statistics
19 # Your code here
20
21 # Display results
22 # Your code here

```

Testing Checklist:

- ☐ Program runs without errors
- ☐ Collects correct number of scores
- ☐ Calculates average correctly
- ☐ Finds min and max correctly
- ☐ Counts high scores correctly
- ☐ Output is formatted clearly

7 Part 7: Reflection Questions (5 points)

Time estimate: 10 minutes

Answer these questions in complete sentences:

1. Why is list indexing starting from 0 confusing, and how do you remember it?

2. When would you use `append()` versus `insert()`? Give an example.

3. How do lists make programs more powerful than using individual variables?

8 Bonus Section: Extra Challenges (Optional - 10 extra points)

Only attempt if you've finished everything else!

8.1 Challenge 1: List Pattern Generator (5 points)

Open [week07_bonus1.py](#) and create a program that generates lists based on patterns:

```

1 # Pattern Generator
2 # Create three different lists:
3
4 # Pattern 1: Even numbers from 2 to 20

```

```

5 even_numbers = []
6 for num in range(____, ____, ____):
7     even_numbers.append(num)
8 print(f"Even numbers: {even_numbers}")
9
10 # Pattern 2: Multiples of 5 from 5 to 50
11 multiples_of_5 = []
12 # Your code here
13
14 print(f"Multiples of 5: {multiples_of_5}")
15
16 # Pattern 3: Squares of numbers 1 to 10 (1, 4, 9, 16...)
17 squares = []
18 # Your code here
19
20 print(f"Squares: {squares}")

```

8.2 Challenge 2: List Menu System (5 points)

Create a program called `list_menu.py` that provides a menu for list operations:

```

=== List Manager ===
1. Add item
2. Remove item
3. Show all items
4. Count items
5. Exit
Choice: 1
Enter item: pizza
Item added!

Choice: 3
Your list: ['pizza']

```

Use a while loop to keep showing the menu until the user chooses Exit!

9 Debugging Discoveries (5 bonus points)

Growth Mindset: Errors are learning opportunities!

Did you encounter any interesting errors while completing this homework? Share one error and how you fixed it:

🔧 My Debugging Story

The Error I Got: _____

What I Tried First: _____

How I Fixed It: _____

What I Learned: _____

No errors? Share a tricky part you figured out instead!

Submission Checklist

Before submitting, make sure you have:

- ☐ Completed all required sections (Parts 1-7)
- ☐ Saved each code exercise in its own file:
 - week07_ex2_1.py, week07_ex2_2.py
 - week07_ex4_1.py, week07_ex4_2.py
 - week07_ex5_1_fixed.py, week07_ex5_2_fixed.py
 - high_score_tracker.py
- ☐ Created a folder named `week07_hw_yourname` with all files
- ☐ Tested each code file individually to ensure it runs without errors
- ☐ Written your name at the top of this paper
- ☐ Answered all reflection questions
- ☐ Attempted bonus section (optional)

Module 2 Concepts Check:

Which list method was most confusing? _____

What do you want to learn next about lists? _____

Time Tracking:

How long did this homework take you? _____ hours

Parent/Guardian Signature: _____

(Confirming student completed work independently)

Great job exploring lists!

Next week: List mutability and why `list2 = list1` doesn't create a copy!